Good To Go

User Manual

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Good To Go shows when your bus, train or tram leaves, in realtime, so you know when to head out the door. This manual is for the person using the device. To build one, see the README.

1 The device

- **Screen:** two rows of 16 characters.
- **Button:** the only control. It is used with clicks and holds, see Everyday use.
- **USB cable:** powers the device. Any USB phone charger works.

You need:

- A **2.4 GHz WiFi network** with a password (WPA2 or WPA3). 5 GHz-only networks don't work.
- A **phone or computer** with a web browser for the setup.
- Two **API keys**, one from Trafiklab and one from Google. If the device came with them built in, you can skip them. Otherwise, ask the person who built it, or see Create the API keys in the README.

2 First-time setup

1. **Plug in the device.** The screen shows `Configuring...` / `Soon good to go!`, then the name and password of the device's own WiFi network:

   ```
   WIFI: GTG12345
   PASS: 12345678
   ```

   The digits are random and change each time the device's network starts.

2. **Join that network** with your phone or computer, using the password on the screen. The screen changes to the address of the setup page and the password to log in to it:

   ```
   goodtogo.local
   PASS: K7MPX2RQ9D
   ```

   This is a different password from the network's. It is made anew every time the setup page is turned on.

3. **Open `https://goodtogo.local`** in your browser.

   Your browser may warn that the connection isn't private. This is expected: the device uses its own certificate, which your browser doesn't know. Choose *Advanced* and continue to the page (the wording differs between browsers). To get rid of the warning, install the device's certificate, see Generate certificates in the README.

4. **Log in.** Type the password from the screen and press **Log in**. Small letters work too. The screen changes to `Configuring...`.

   You stay logged in while you use the page. After 5 minutes without using it, you are asked to log in again.

   Above the steps, **Device health** shows what the device has, each with a check mark when done and a cross when not, see Device health. The setup has to be finished in order before anything else.

5. **Connect to your WiFi.**

   1. Press **Scan for networks** and choose your home network, or type its name.
   2. Enter the password and press **Connect**.

   Once the device is connected, the page moves on to the API keys. When you later leave the device's network, it shuts down about 5 seconds later, and the page is reachable at `https://goodtogo.local` on your home network instead. Your phone may drop off the device's network for a moment when the device joins yours: it rejoins by itself.

6. **Add the API keys.** Enter the Trafiklab key and the Google key and press **Verify** for each. The device checks each key with its service, and the page shows a check mark and *Key verified* when it works. Keys built into the device are verified by themselves and shown as verified.

7. **Add your departures.** You can track up to 8.
    1. Type the name of your stop, e.g. *Sveavägen*, and press **Search**. The 10 best matches are listed with their town, so stops with the same name can be told apart.
    2. Choose your stop and press **Search for available departures**. This looks through a whole day of departures and can take up to a minute.
    3. Choose a line and direction and press **Add to departures**. Departures you already track are marked *Tracked*.
    4. Repeat for every departure you want. Your saved departures are listed at the top of the page, and the cross next to one removes it.
8. **Press Done.** The page says *Your device is good to go!* when everything is set up, or what is still missing. Once set up, the page switches to the settings, see Changing the setup.

The screen then shows `U R Good To Go!` / `Press to search.` The setup page stays on, and the screen shows `Stop config mode` / `Hold 5 sec...` until you hold the button for 5 seconds, or until the page has been unused for 5 minutes.

3 Everyday use

3.1 The button

Push	What happens
Click	Shows the next departure. Fetches new times first if they are a minute or older.
Double-click	Toggles the top row between the line and destination and the stop name.
Five-click	Puts the device to sleep, see Sleep.
Hold 5 seconds	Turns the setup page on or off.
Hold 20 seconds	Factory reset, see Factory reset.

A click shows its departure after a short moment, as the device first checks that no more clicks follow. Two quick clicks are a double-click, and a five-click is five quick clicks. Three or four quick clicks do nothing.

A double-click only works while a departure is shown. Each departure starts with its line and destination on the top row, a double-click shows its stop name instead, and the next double-click shows the line and destination again. A click to the next departure always starts with the line and destination.

The holds happen while you still hold the button: at 5 seconds the setup page turns on or off, and you can let go. Keep holding to 20 seconds and the factory reset starts.

While the setup page is on, only the holds work. Clicks, double-clicks and five-clicks are ignored, so the password on the screen stays there. Hold the button for 5 seconds to turn the setup page off, then click as usual.

3.2 Reading a departure

```
515 Sundbybergs.
now, 13, 28 min
```

- **Top row:** the line and its destination. A name too long for the screen is cut, and ends with a full stop.

- **Bottom row:** the minutes until the next 3 departures. `now` means less than a minute.

The times count down by themselves, and new times are fetched in the background every minute while a departure is shown. Realtime estimates are used when the operator has them, otherwise the timetable. Cancelled departures and departures more than 60 minutes away are left out.

The first click after `U R Good To Go!` shows the first departure. After the last departure, a click shows `Need more stops?` / `Hold 5 sec...`, and the next one starts over from the first.

The screen has no å, ä or ö, so they are shown as a, a and o.

4 Changing the setup

1. Hold the button for 5 seconds, until the screen changes.
2. If the device is connected to your WiFi, the screen shows `goodtogo.local` and the login password. Open `https://goodtogo.local` from a phone or computer on the same network and log in with that password.

   If it isn't connected, the screen shows the device's own network and password instead. Join it, and the screen shows the login password. Then open the page as in First-time setup.
3. Make your changes in the settings, see below.
4. Hold the button for 5 seconds again to turn the page off, or leave it: it turns off by itself after 5 minutes without use.

If the device isn't fully set up, e.g. after its WiFi was lost, the page shows the setup steps instead, from the first one that isn't done.

4.1 The settings

A set-up device shows Device health and three sections. Each section has a check mark or a cross in its title, and opens when you tap it:

- **WiFi:** shows the network the device is connected to. To change it, press **Scan for networks**, choose the network, enter its password and press **Connect**.
  - The device leaves its current network while it joins the new one.
  - If your phone or computer isn't on the new network, the page loses the device: join the new network and open `https://goodtogo.local` again.
  - If the device can't join the new network, e.g. because of a wrong password, it opens its own network and shows its name and password on the screen. Join it, open the page and try again. Restarting the device instead, e.g. by waking it from sleep, takes it back to the old network.
- **API keys:** enter a new Trafiklab or Google key and press **Verify**. A key that is rejected doesn't replace the one the device has.
- **Departures:** your saved departures, with a cross to remove one, and the stop search to add more, as in the setup. Changes take effect right away, and the device starts over from the first departure.

Done at the bottom asks the device whether everything is set up. Once it is, the screen shows `Stop config mode` / `Hold 5 sec...`.

4.2 Device health

Check	Has a check mark when
WiFi connected	The device is connected to your WiFi.
Clock synced	The device has the time from the internet.
API keys verified	Both keys have been accepted by Trafiklab and Google.
Departures saved	At least one departure is saved.
Good to go	The device is connected, has working keys and has departures, so it can show them.

The checks are updated after each change you make, and when you come back to the page. A cross shows what to fix: open that section in the settings.

5 Sleep

After **5 minutes** without a button push, or without use of the setup page, the device shows `Going to sleep` / `Press to wake` and turns the screen off.

Click the button five times quickly to put it to sleep at once, e.g. when you leave home. This doesn't work while the setup page is on: turn it off first by holding the button for 5 seconds.

Push the button once to wake it. The screen shows `Waking up...` while it reconnects to your WiFi, then it fetches all departures (`Fetching...`) and shows `U R Good To Go!`. A wake is quicker than turning the device on: it remembers what was set up before it went to sleep, so it doesn't check the keys or wait for the clock again.

If something wasn't set up when it went to sleep, e.g. it fell asleep on the setup page, it wakes straight into the setup page at that step. After a power cut it starts as when turned on, checking everything.

6 Messages on the screen

The departure messages on the bottom row keep the departure's name on the top row.

Top row	Bottom row	Meaning
`Configuring...`	`Soon good to go!`	Starting up.
`Waking up...`	`Soon good to go!`	Waking from sleep, reconnecting to the WiFi.
`WIFI: GTG...`	`PASS: ...`	The setup page is on and the device has its own network. Join it to set it up.
`goodtogo.local`	`PASS: ...`	Open `https://goodtogo.local` and log in with this password, which is new each time the setup page is turned on.
`Configuring...`		Someone logged in to the setup page.
`Fetching...`		Fetching all departures after starting.
`U R Good To Go!`	`Press to search.`	Ready. Click to see a departure.
	`Updating...`	Fetching new times for the shown departure.
	`No departures...`	Nothing leaves within 60 minutes.
	`Fetch failed`	The times couldn't be fetched. It is tried again in a minute or on the next click.
`Need more stops?`	`Hold 5 sec...`	After the last departure. Hold the button to add more on the setup page.

Top row	Bottom row	Meaning
`Stop config mode`	`Hold 5 sec...`	The setup page is on. Hold the button for 5 seconds to turn it off.
`Connection lost!`	`Retrying: 2/5`	The WiFi dropped, the device is trying to reconnect.
`Connection lost!`	`Starting config`	Reconnecting failed, the setup page starts. See Troubleshooting.
`Fetch failed 3x`	`Starting config`	Three fetches in a row failed, the setup page starts.
`Error:`	`No connection...`	The device isn't connected to WiFi.
`Error:`	`Time unsynched`	The device doesn't know the time yet. It needs internet access.
`Error:`	`API keys invalid`	A key is missing or was rejected. Enter it again on the setup page.
`Error:`	`No stops added`	No departures are saved. Add them on the setup page.
`Going to sleep`	`Press to wake`	Going to sleep, push the button to wake it.
`Factory reset`	`Restarting...`	Everything is erased, the device restarts into the setup.
`Factory reset`	`failed, retry`	The reset failed. Hold the button for 20 seconds again.

7 Troubleshooting

7.1 The WiFi connection was lost

The device tries to reconnect 5 times (`Retrying: 1/5` to `5/5`). If it succeeds, it goes back to what it showed. If not, the setup page starts with the device's own network. When your WiFi is back, either:

- push the button to wake the device once it has gone to sleep, which restarts it and reconnects, or
- join the device's network, open `https://goodtogo.local` and connect to your WiFi again.

7.2 Fetch failed 3x

The device has WiFi but can't get departures. Your internet may be down, or Trafiklab may be unavailable. Check the API keys on the setup page, and wait a few minutes before trying again. A fetch that works resets the count.

7.3 The page keeps asking me to log in

- You are logged out after 5 minutes without using the page, and when the setup page is turned off. Log in again with the password on the screen.
- If the screen no longer shows the password, e.g. because someone already logged in, hold the button for 5 seconds twice: once to turn the setup page off and once to turn it on. The screen then shows a new password.
- The browser must allow cookies for `goodtogo.local`, as that is how it stays logged in.

7.4 Changing the WiFi failed

The device couldn't join the network you chose and opened its own network. Its name and password are on the screen. Join it, open `https://goodtogo.local`, log in and try again: check the network's name and password, and that it is a 2.4 GHz network. To go back to the old network instead, push the button to wake the device once it has gone to sleep, which restarts it.

7.5 Wrong password

Check the password on the screen again: it has no I, O, 0 or 1, so a character that looks like one of them is another one. After 5 wrong passwords in a row, the page asks you to wait 30 seconds before trying again.

7.6 goodtogo.local doesn't open

- Make sure your phone or computer is on the device's network, or on the same home network as the device.
- Type the whole address, including `https://`.
- Check that the screen shows that the setup page is on. If not, hold the button for 5 seconds.
- Turn off mobile data on your phone while joined to the device's network, so the phone doesn't use mobile data instead.

7.7 My WiFi network isn't in the list

The device only sees 2.4 GHz networks, and hidden networks aren't listed. Networks without a password are listed, marked *open*, but the device can't connect to them.

7.8 The screen is blank

The device may be asleep: push the button once. If it stays blank, check the USB power.

7.9 The screen lights up but shows no text

The contrast needs adjusting, with the small screw on the back of the screen.

7.10 I've changed WiFi password or moved

Hold the button for 5 seconds. If the device can't connect, it shows its own network: join it and connect to the new WiFi from the setup page.

8 Factory reset

A factory reset erases the WiFi settings, the API keys you entered and all saved departures. API keys built into the device stay, and are used again after the reset.

Hold the button for 20 seconds. At 5 seconds the setup page turns on or off as usual: keep holding. At 20 seconds the screen shows `Factory reset` / `Restarting...` and the device restarts into First-time setup. This works at any time, also during startup.

9 Privacy and security

- **The setup page needs the password on the device's screen**, so only someone who can see the device can log in. The password changes every time the setup page is turned on. The page turns off 5 minutes after it was last used, or when you hold the button for 5 seconds.
- **Install the device's certificate** if you can, see Generate certificates in the README. Without it your browser can't tell the device from someone imitating it on your network, who could then catch the login password.
- **The WiFi password and API keys are stored unencrypted** on the device. Do a factory reset before you give the device away.
- The device only contacts Trafiklab and Google for departures and stop towns, and time servers for the clock. Both services are reached over encrypted connections.
- **Use a strong WiFi password.** The device can be tricked into using the older WPA2 standard by someone imitating your network, which makes a weak password easier to guess.